

Polymarket Exploratory Data Analysis (EDA) Report

Executive Summary

The dataset given has one row per Polymarket trader and it tells us their profit, how much they traded, how often they traded, which news topics they bet on, and a skill tag of awful, bad, good, or sharp. This report explores why the data appears the way it does. This EDA template is borrowed from github user [SarahAA1](#). This EDA was conducted using Python and pandas.

Data Overview

Dataset Information

- **Number of Traders (Rows):** 604,578 (no duplicates)
- **Number of Columns:** 41 (9 activity/profit metrics, 6 execution quality metric, 17 topic share columns and 1 label column)
- **Net PnL:** \$0 (consistent with Polymarket being a zero sum market)

Data Loading and Cleaning

Data was loaded with `pandas.read_parquet` and is clean. There are no duplicate rows and no duplicate trade ID's.

Data Summary and Descriptive Statistics

Summary Statistics

Statistics for the core activity and profitability columns:

Table 1. Summary statistics for core activity and profitability columns.

	count	mean	std	min	25%	50%	75%	max
trader_pnl	604,578	0.000	6,471.315	-1,219,869.872	-7.527	-0.053	0.799	1,103,256
trader_volume	604,578	11,139.970	311,389.933	0.010	36.226	148.350	754.000	91,579,410
transaction_count	604,578	43.468	1,142.982	1.000	3.000	7.000	16.000	4,194,990
transactions_per_day	604,578	3.863	14.938	1.000	1.167	2.000	3.267	4,805.647
mean_tx_value	604,578	54.661	546.487	0.000	2.700	8.451	27.857	110,230
trader_ppv	604,578	-0.025	0.170	-0.999	-0.043	-0.000	0.013	0.999

trader_volume and mean_tx_value columns are right skewed with the signature of a long tail of large traders

pulling the average up. trader_pnl and trader_ppv are more centered near zero, an expected outcome in a zero sum market.

Feature Distribution

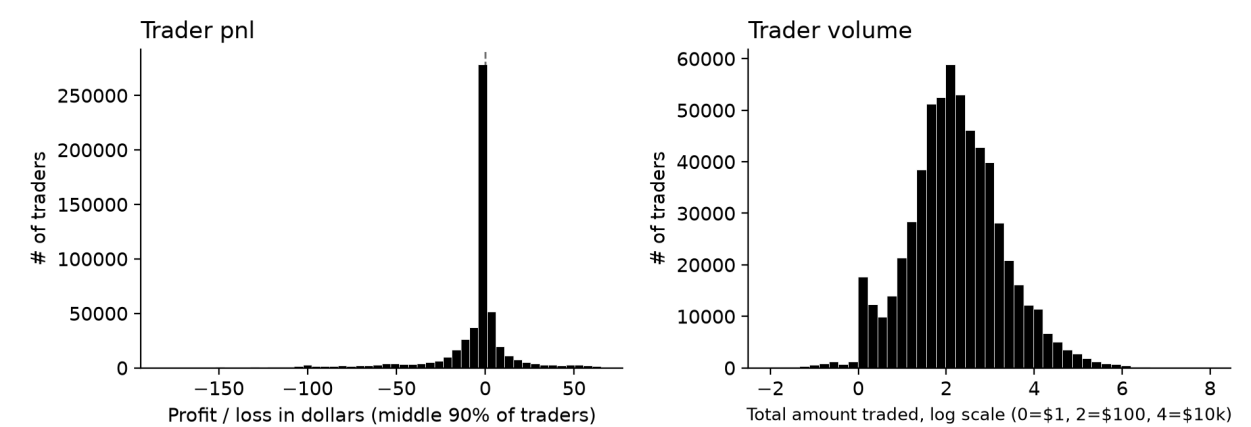


Figure 1. Distribution of trader PnL (middle 90% of traders) and trader volume (log scale).

PnL Distribution zoomed in at the middle 90% to avoid the huge spike at \$0. Log scale is used for trader volume because it spans many orders of magnitude.

Missing Values Analysis

4 of the 41 columns have missing or invalid values.

Table 2. Columns with missing values.

Column	missing_count	missing_pct	Missing rows that are 1-trade traders
std_delta	75,855	12.55	100.0%
std_time	75,855	12.55	100.0%
std_tx_value	75,855	12.55	100.0%
std_time_vw	4,156	0.69	-

Columns with infinite values: std_time_vw (1).

All four affected columns are standard deviation metrics and these gaps all occur because it's not possible to compute standard deviation from only one trade. Handling strategy: leave these as missing (NaN) since filling it with 0 or a mean would make the single trade traders look artificially consistent. The infinite value in std_time_vw was dropped before any aggregate calculation.

Bivariate Analysis

Numeric-Numeric Relationships

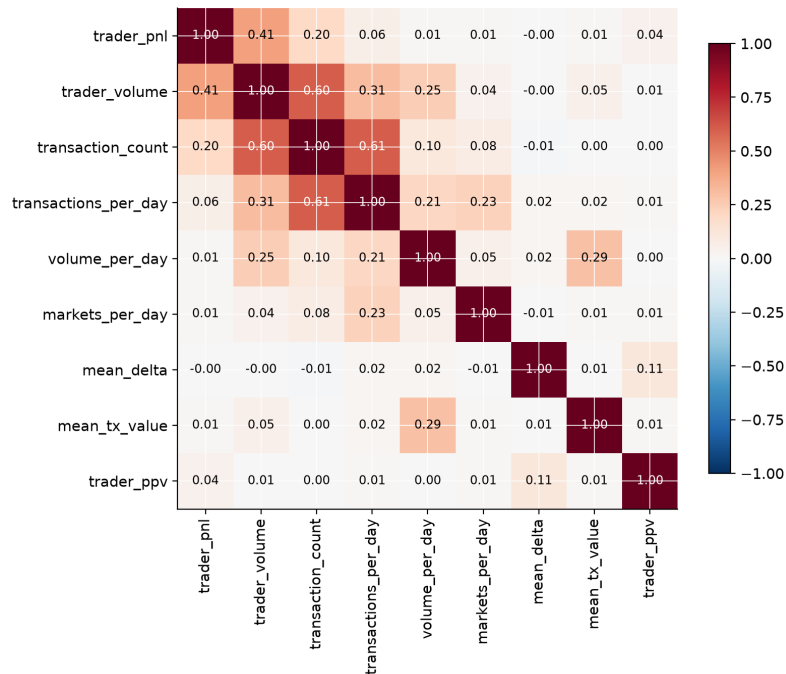


Figure 2. Correlation matrix of activity and profitability metrics.

Most relationships are weak and the clearest ones are not too surprising. transaction_count correlates 0.60 with trader_volume which makes sense as trading more times tends to move more total dollars.

Transaction_count correlates 0.61 with transactions_per_day which also makes sense as more trades usually mean a faster pace, not just more history. trader_pnl correlated 0.41 with trader_volume as bigger traders tend to have bigger absolutely PnL swings. trader_ppv barely correlates with size at all which tells us that size and trading skill are separate things.

Categorical-Numeric Relationships

Table 3. Average and median outcomes by trader_label.

trader_label	Mean PnL	Median PnL	Mean Volume
awful	-476.37	-17.17	3,003.82
bad	-180.53	-1.67	11,968.29
good	226.71	0.26	19,449.13
sharp	550.14	12.68	4,728.56

Average and median outcomes by skill move as expected with awful traders losing the most and sharp traders winning the most.

Outlier Detection

trader_volume and trader_pnl both have extreme outliers. There is a \$91.6M volume trader against a \$148 median and there is a \$1.1M single trader profit swing. This observation demonstrates how concentrated the platform's activity is.

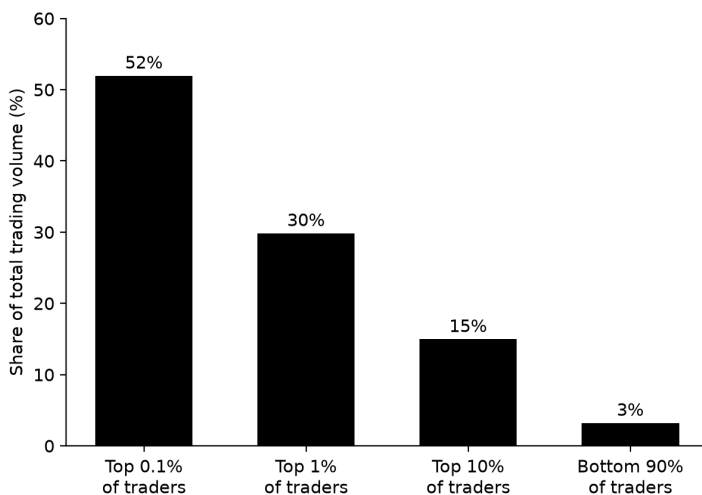


Figure 3. Share of total trading volume by trader size group.

Handling strategy: these outliers were kept in every calculation since dropping the top 0.1% (about 600 accounts) of traders would mean throwing out over half the platform's trading volume. In charts where the shape would be unreadable because of these outliers, values were clipped for readability. Every summary statistic and correlation elsewhere still uses the full, uncapped data.

Custom Analyses

Execution Style by Skill Level

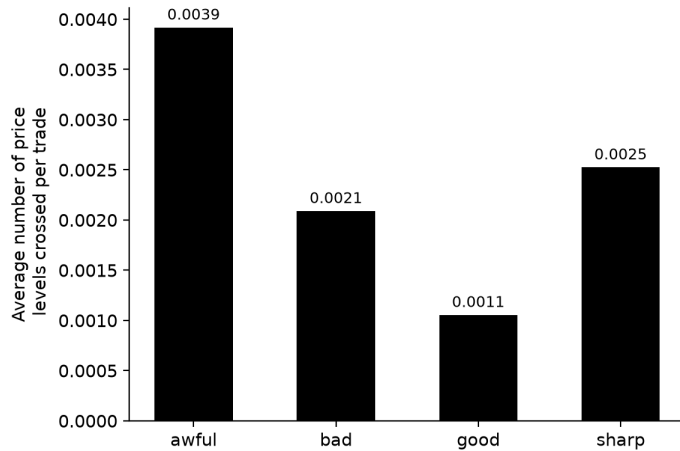


Figure 4. Average price levels crossed per trade by skill level.

The best and the worst traders trade alike. `price_levels_per_transaction` measures how much of the order book a trader typically has to cross to get filled, essentially how aggressive they trade. Notably, sharp and awful traders are more aggressive than bad and good traders. This tells me that in order to incur profit, one must be willing to take risks. However, taking on such aggressive trading also does not guarantee profit. Bad and good traders are more passive and therefore less likely to produce extreme results.

Topic Exposure

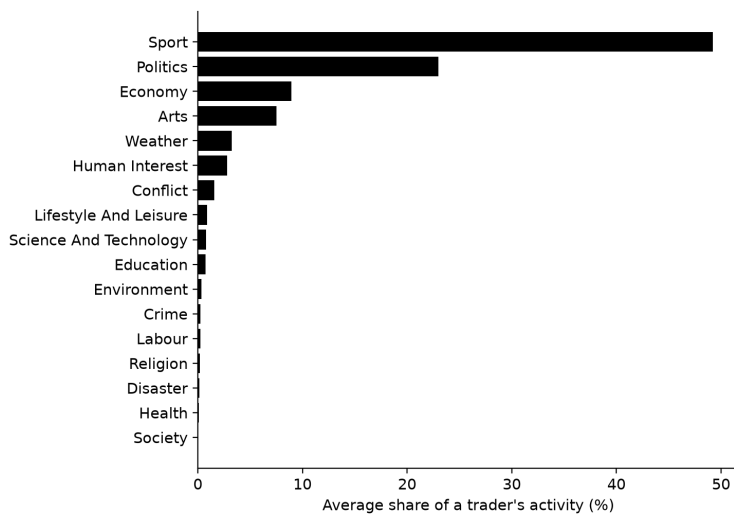


Figure 5. Average share of a trader's activity by news topic.

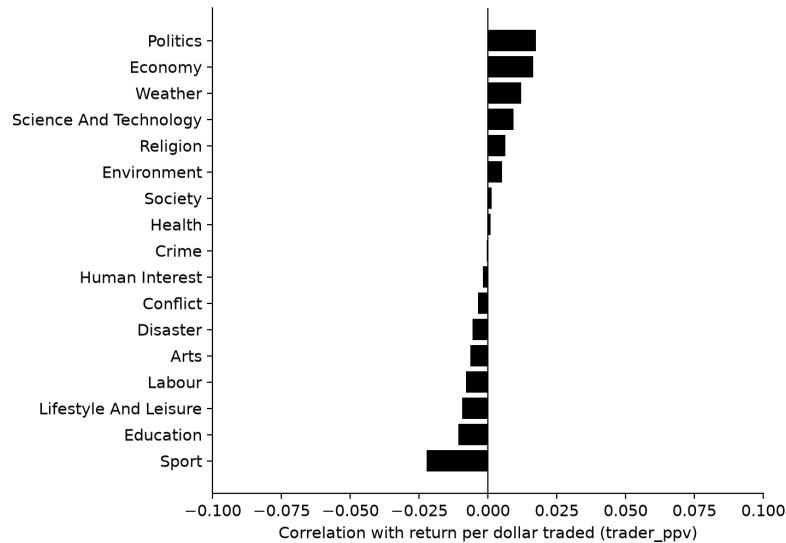


Figure 6. Correlation of each topic's share with return per dollar traded (trader_ppv).

Each trader's activity is broken down across 17 news topics. Checking each topic against its profitability shows a very weak relationship which tells us that what a trader bets on barely matters.

Conclusions

This dataset describes a zero sum market, consistent with Polymarket being a zero sum market, where a small group of traders account for most of the volume and money at stake. Trading more aggressively seems to be a trend among the most extreme (awful and sharp) of traders and there seems to be no correlation between what is bet on and a traders profit.

Next Steps

Since the top 0.1% dominate most of the volume, an average trader would mostly describe a few hundred large accounts. Next steps would be to analyze large and small traders separately so that the average trader does not mix the two populations. It'd be good to analyze the next year's market as well to see if these trends still hold true or if something has changed or betting on one topic has become more profitable.